Abdelrahman Khalifa

Software Engineer | C++, Multithreading, Database Systems, Linux

abdelrahman.khalifa.gad@gmail.com | +20 114 250 5042 | Cairo, Egypt

linkedin.com/in/abd002 | github.com/Abd002 | codeforces.com/profile/_Khalifa

SUMMARY

Systems software engineer working in modern C++, Rust and C on Linux. Studying database internals through Carnegie Mellon's database systems course (CMU 15-445: storage, buffer pool, storage models, hash tables; a thread-safe C++ project that passes the course autograder) and distributed systems through the Raft paper and *Designing Data-Intensive Applications*. Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting), with 11 PRs merged upstream. Codeforces Expert and 4-time ECPC finalist.

EXPERIENCE

Google Summer of Code Contributor | **The Linux Foundation (OpenPrinting)** | **May 2026 – Oct 2026**

Open-source project: Printers page for COSMIC Settings, in Rust; passed the final evaluation | github.com/Abd002/GSOC-2026

- Built the **Printers page for COSMIC Settings** and a standalone app with the same UI: printer status and supply levels, print queues, live updates and an Add Printer flow (cosmic-settings #2182, in review).
- Designed it as 5 crates with an async client and ran the C FFI backend as a separate varlink IPC service, so a crash in native code cannot take down the Settings app.
- Got **11 PRs merged upstream** (cups-rs, #164, #165): ported cups-rs to libcups3 with a CI matrix across architectures and CUPS versions, added DNS-SD bindings, and fixed crashes and a leaked client in libcups (C).

Embedded Software Engineer | **EKSON** | **Aug 2025 – Present**

Software for 3-DoF and 6-DoF pneumatic motion simulators: firmware, desktop app, backend and CI

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Cut the desktop control app's CPU usage and latency by about 40% (measured) by waiting for ESP32 and server messages concurrently with asyncio, off the UI thread.

Embedded Linux Systems Intern | **Orange Digital Center (ODC)** | **Dec 2024 – Jan 2025**

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OPEN SOURCE PROJECTS

Contributor (all merged) — Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | **Mar 2025 – Dec 2025**

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553); Rust Clippy: fixed `ref_as_ptr` false positives (PR #16214).

PROJECTS

- **Database Systems, CMU 15-445** (C++17): thread-safe Count-Min Sketch for the BusTub database: a depth × width counter matrix with one `std::mutex` per row so inserts on different rows run in parallel; count, merge, top-k and move semantics. SQL analytics (joins, aggregation, upserts) in DuckDB. GitHub
- **Kernel Monitor** (C, Linux, ARM, QEMU): character device driver that walks the process list under RCU and exports per-task IDs, memory, threads, context switches and CPU time; a pthread user-space reader polls it. GitHub
- **Operating System Simulator** (Java, team of 3): built the kernel, system calls and process model of a CLI OS with round-robin scheduling, memory management and mutexes. GitHub
- **IoT Device Communication** (C++, Linux, QEMU): TCP client–server classes; clients cross-compiled for ARM and run on two QEMU boards over a bridged network. GitHub

TECHNICAL SKILLS

Languages: Modern C++ (C++17, STL), Rust, C, Python, SQL, Java, Bash

Computer science: Data structures and algorithms, concurrency and multithreading (mutexes, RCU, async), operating systems, memory management, networking (TCP/UDP, IPC)

Databases & distributed: Database internals (storage and page layout, buffer pool, row/column storage and compression, hash tables), SQL, DuckDB, MongoDB; Raft, replication and consistency models

Linux & tools: Linux (Ubuntu/Debian, Arch), kernel modules, FFI, IPC (varlink); Git, CMake, Make, Cargo, GitHub Actions CI, testing, debugging, code review

ACHIEVEMENTS

- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024; mentor in the local ICPC community, teaching problem solving in C++ to 20+ colleagues.

EDUCATION

Bachelor of Science (B.Sc.), Computer and Systems Engineering | **Minia University** | **2020 – 2025**

- Grade: Very Good with Honors. Treasurer, IEEE Student Branch (budget of more than $10k).